



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0510 – COMPUTER VISION

Name: Premarubha N

EXPERIMENT NO: 25

Reg No: 192525020

25. Perform Sharpening of Image using Gradient masking.

AIM:

Sharpening of Image using Gradient masking

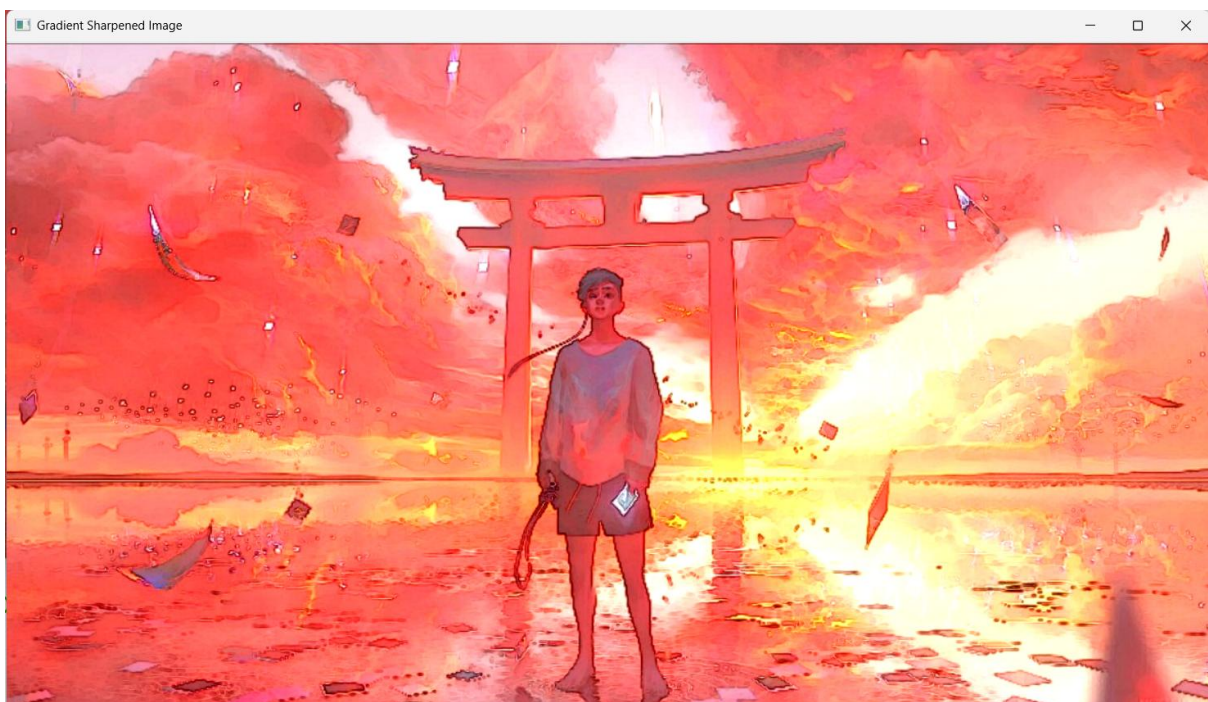
PROGRAM:

```
import cv2
import numpy as np
img = cv2.imread(r"C:\Users\premarubha\Desktop\cv model image.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
grad_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
grad_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
gradient = cv2.magnitude(grad_x, grad_y)
gradient = cv2.convertScaleAbs(gradient)
gradient = cv2.cvtColor(gradient, cv2.COLOR_GRAY2BGR)
sharpened = cv2.addWeighted(img, 1.5, gradient, -0.5, 0)
cv2.imshow("Original Image", img)
cv2.imshow("Gradient Sharpened Image", sharpened)
cv2.imwrite(r"C:\Users\premarubha\Desktop\gradient_sharpened_image.jpg", sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.